

Operations in Linear Array with implementation in C

Topic 2

Array Initialization
Accessing Elements
Insertion
Deletion
Searching
Updating

Learning Objectives

By the end of this chapter, you will be able to:

- Discuss the Linear Array Initialization
- Explain the Concepts of Accessing Elements in Linear Array
- Explain the Concepts of Element Insertion, Deletion, Searching, Updating in Linear Array



Linear Array in C

A linear array is a collection of elements of the same data type arranged in a sequential manner. Each element in the array can be accessed using its index, which starts from 0 for the first element and goes up to the size of the array minus 1 for the last element.

Let's cover the following operations:

- ☐ Array Initialization
- ☐ Accessing Elements
- ☐ Insertion
- ☐ Deletion
- ☐ Searching
- ☐ Updating

Array Initialization

```
#include <stdio.h>
#include<conio.h>
#define MAX_SIZE 100
void main() {
    int arr[MAX_SIZE]; // Declaration of the array
    int size; // Number of elements in the array
    printf("Enter the size of the array: ");
    scanf("%d", &size);
    printf("Enter the elements of the array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Array initialized successfully!\n");
    getch();
}
```

Array Initialization

```
#include <stdio.h>
#include <conio.h>
#define MAX_SIZE 100
void main() {
    int arr[MAX_SIZE]; // Declaration of the array
    int size; // Number of elements in the array
    printf("Enter the size of the array: ");
    scanf("%d", &size);
    printf("Enter the elements of the array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Array initialized successfully!\n");
    getch();
}
```

Accessing Elements

```
#include <stdio.h>

#define MAX_SIZE 100

void main() {
    int arr[MAX_SIZE];

    int size, index;

    printf("Enter the size of the array: ");

    scanf("%d", &size);

    printf("Enter the elements of the array:\n");

    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
```

```
printf("Enter the index of the element you want to access: ");

scanf("%d", &index);

if (index >= 0 && index < size) {
    printf("Element at index %d: %d\n", index, arr[index]);
}

else {
    printf("Invalid index! Please enter an index between 0 and %d.\n", size - 1);
}

getch();
}
```

Insertion

```
#include <stdio.h>
#define MAX_SIZE 100
int main() {
    int arr[MAX_SIZE];
    int size, index, value;
    printf("Enter the size of the array: ");
    scanf("%d", &size);
    printf("Enter the elements of the array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the index where you want to
           insert the element: ");
    scanf("%d", &index);
    printf("Enter the value to be inserted: ");
    scanf("%d", &value);
```

```
    if (index < 0 || index > size) {
        printf ("Invalid index! Please enter an index
                between 0 and %d.\n", size);
    }
    else {
        // Shift elements to the right to make space for
        the new element
        for (int i = size - 1; i >= index; i--) {
            arr[i + 1] = arr[i];
        }
        arr[index] = value;
        size++; // Increase the size of the array
        printf("Element inserted successfully!\n");
    }
    getch();
}
```

Deletion

```
#include <stdio.h>
#define MAX_SIZE 100

void main() {
    int arr[MAX_SIZE];
    int size, index;
    printf("Enter the size of the array: ");
    scanf("%d", &size);
    printf("Enter the elements of the array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the index of the element you
want to delete: ");
    scanf("%d", &index);
```

```
    if (index < 0 || index >= size) {
        printf("Invalid index! Please enter an index between 0
and %d.\n", size - 1);
    }
    else {
        // Shift elements to the left to overwrite the element
to be deleted
        for (int i = index; i < size - 1; i++) {
            arr[i] = arr[i + 1];
        }
        size--; // Decrease the size of the array
        printf("Element deleted successfully!\n");
    }
    getch();
}
```


Searching

```
#include <stdio.h>

#define MAX_SIZE 100

int main() {
    void arr[MAX_SIZE];
    int size, value;
    printf("Enter the size of the array: ");
    scanf("%d", &size);
    printf("Enter the elements of the array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the value to be searched: ");
    scanf("%d", &value);
```

```
    int found = 0;
    int index = -1;
    for (int i = 0; i < size; i++) {
        if (arr[i] == value)
        {
            found = 1;
            index = i;
            break;
        }
    }
    if (found) {
        printf("Value found at index %d.\n", index);
    }
    else {
        printf("Value not found in the array.\n");
    }
    getch();
}
```

Updating

```
#include <stdio.h>

#define MAX_SIZE 100

void main() {
    int arr[MAX_SIZE];
    int size, index, value;
    printf("Enter the size of the array: ");
    scanf("%d", &size);
    printf("Enter the elements of the array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }
```

```
    printf("Enter the index of the element you want to update: ");
    scanf("%d", &index);

    if (index >= 0 && index < size) {
        printf("Enter the new value: ");
        scanf("%d", &value);
        arr[index] = value;
        printf("Element updated successfully!\n");
    }
    else {
        printf("Invalid index! Please enter an index between 0 and
%d.\n", size - 1);
    }
    getch();
}
```

THANK YOU